Z-Scores & Linear Transformations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Computing z-scores**

A test has mean 70 and standard deviation 5. Find the z-score for a student who scored 82.

$z = \frac{x - \mu}{\sigma} = \frac{82 - 70}{5} = \frac{12}{5} = 2.4.$

2 **[Foundational]** • **Target: Computing z-scores**

Explain in words what a z-score of -1.5 means, in the context of a data value's position relative to its distribution.

A z-score of -1.5 means the data value lies 1.5 standard deviations below the mean of its distribution.

3 **[Foundational]** • **Target: Comparing values across different distributions**

On Test A (mean 75, standard deviation 6), a student scores 87. On Test B (mean 70, standard deviation 4), a different student scores 77. Compare the two students' relative performance using z-scores.

Test A: $z = \frac{87 - 75}{6} = \frac{12}{6} = 2$. Test B: $z = \frac{77 - 70}{4} = \frac{7}{4} = 1.75$. Since $2 > 1.75$, the student on Test A performed better relative to their own test's distribution, even though the raw score gap (12 vs. 7) might suggest otherwise.

4 **[Foundational]** • **Target: Effect of adding a constant to every data value**

A data set has mean 50 and standard deviation 8. If 10 is added to every value, find the new mean and new standard deviation.

Adding a constant shifts the mean by that constant but does not change the spread: new mean $= 50 + 10 = 60$; standard deviation stays 8.

5 **[Foundational]** • **Target: Effect of adding a constant to every data value**

Explain, without doing a full calculation, why adding a constant to every data value does not change the standard deviation.

Standard deviation measures how spread out the data is around the mean. Adding the same constant to every value shifts the entire data set (and its mean) by that constant, but every value's distance from the (also shifted) mean stays exactly the same — so the spread, and hence the standard deviation, is unaffected.

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6 [Foundational] • Target: Effect of multiplying every data value by a constant

A data set has mean 20 and standard deviation 3. If every value is multiplied by 4, find the new mean and new standard deviation.

Multiplying by a constant scales both the mean and the standard deviation by that constant (standard deviation uses the absolute value of the constant): new mean $= 20 \times 4 = 80$; new standard deviation $= 3 \times 4 = 12$.

7 [Foundational] • Target: Effect of multiplying every data value by a constant

A distance measured in miles has mean 10 and standard deviation 2. Convert both to kilometers (multiply by 1.609), and find the new mean and standard deviation, to two decimal places.

New mean $= 10 \times 1.609 = 16.09$ km. New standard deviation $= 2 \times 1.609 = 3.22$ km.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Visualizing a standardized (z-score) distribution

The graph shows the standard normal distribution, which has mean 0 and standard deviation 1 — the distribution that results after standardizing any normal data set using z-scores.

a State the mean and standard deviation shown by this curve. Mean $= 0$ (the curve is centered at $x = 0$) and standard deviation $= 1$ (this is the standard normal curve).

b Explain why standardizing a data set using $z = \frac{x - \mu}{\sigma}$ always produces a distribution with mean 0 and standard deviation 1, regardless of the original data's mean and standard deviation. Subtracting μ shifts the mean to 0 (since subtracting a constant equal to the mean re-centers the data there), and dividing by σ rescales the standard deviation to $\frac{\sigma}{\sigma} = 1$ — combining a shift and a scaling that always produces mean 0, standard deviation 1.

(a) Mean $= 0$ (the curve is centered at $x = 0$) and standard deviation $= 1$ (this is the standard normal curve).; (b) Subtracting μ shifts the mean to 0 (since subtracting a constant equal to the mean re-centers the data there), and dividing by σ rescales the standard deviation to $\frac{\sigma}{\sigma} = 1$ — combining a shift and a scaling that always produces mean 0, standard deviation 1.

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9 **[Intermediate]** • **Target: Applying linear transformations in context**

A factory's daily output (in units) has mean 500 and standard deviation 30. Management adds a fixed bonus batch of 50 units to every day's count, then reports output in dozens (divide by 12). Find the new mean and standard deviation, to two decimal places.

Adding 50: new mean $= 550$, standard deviation stays 30. Dividing by 12: new mean $= \frac{550}{12} \approx 45.83$ dozens, new standard deviation $= \frac{30}{12} = 2.5$ dozens.

10 **[Intermediate]** • **Target: Finding a raw value from a z-score**

A distribution has mean $\mu = 64$ and standard deviation $\sigma = 5$. Find the data value corresponding to a z-score of $z = -1.8$.

$x = \mu + z\sigma = 64 + (-1.8)(5) = 64 - 9 = 55$.

11 **[Intermediate]** • **Target: Finding a raw value from a z-score**

A distribution has mean $\mu = 200$ and standard deviation $\sigma = 15$. If a data value has $z = 2.2$, find the data value.

$x = \mu + z\sigma = 200 + 2.2(15) = 200 + 33 = 233$.

12 **[Intermediate]** • **Target: Combined shift-and-scale transformations**

A teacher curves test scores (mean 65, standard deviation 10) by first adding 5 points to every score, then multiplying the result by 1.1. Find the new mean and new standard deviation, to two decimal places.

After adding 5: mean $= 70$, standard deviation stays 10. After multiplying by 1.1: new mean $= 70 \times 1.1 = 77$, new standard deviation $= 10 \times 1.1 = 11$.

13 **[Intermediate]** • **Target: Combined shift-and-scale transformations**

A weight distribution has mean $\mu = 68$ kg and standard deviation $\sigma = 9$ kg. Find the z-scores for weights of 77 kg and 50 kg.

$z_{77} = \frac{77 - 68}{9} = \frac{9}{9} = 1$. $z_{50} = \frac{50 - 68}{9} = \frac{-18}{9} = -2$.

14 **[Intermediate]** • **Target: Combined shift-and-scale transformations**

A national math exam has mean 500 and standard deviation 100, while a national verbal exam has mean 500 and standard deviation 110. A student scores 640 on the math exam and 625 on the verbal exam. Even though the verbal raw score is lower, use z-scores to determine which exam the student performed better on relative to other test-takers, and explain your reasoning.

Math z-score: $\frac{640 - 500}{100} = 1.4$. Verbal z-score: $\frac{625 - 500}{110} \approx 1.14$. Since $1.4 > 1.14$, the student performed better relative to other test-takers on the math exam, despite having a numerically higher raw score on the verbal exam — this shows why comparing raw scores directly across different distributions can be misleading, and z-scores put both exams on a common, comparable scale.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Combined shift-and-scale transformations** • **Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions**

A shipping company records package weights with mean 18 kg and standard deviation 2.5 kg. The company then converts every measurement from kilograms to pounds by multiplying by 2.20462, and separately subtracts 1 kg from every package to account for packaging weight before converting. Describe, step by step, how each operation (the subtraction, then the multiplication) affects the mean and standard deviation.

Subtracting 1 kg from every package: new mean $= 18 - 1 = 17$ kg; standard deviation is unchanged at 2.5 kg (subtracting a constant doesn't affect spread). Multiplying by 2.20462 to convert to pounds: new mean $= 17 \times 2.20462 \approx 37.48$ lbs; new standard deviation $= 2.5 \times 2.20462 \approx 5.51$ lbs (multiplying scales both the mean and the standard deviation by the same factor).

16 **[Challenging]** • **Target: Combined shift-and-scale transformations** • **Cross-Lesson: 3.03 The Normal Distribution**

Level 3 · AdvancedA scatterplot displays a point at $x = 18$, where the rest of the sample data has x-values between 2 and 8. The point falls almost exactly along the linear trend established by the other points. Explain the effect of this point on: (a) leverage, (b) the residual, (c) the correlation coefficient r, and (d) the slope b_1 of the LSRL.

(a) Leverage: The point has high leverage because its x-value (18) is substantially far from the mean $\bar{x}$ of the remaining data. (b) Residual: The point has a very small residual because it falls directly on the linear trajectory established by the other data points. (c) Correlation r: Because the point has an extreme x -value and small residual in line with the trend, it substantially strengthens (increases) the correlation coefficient r toward 1. (d) Slope b_1: The point is not influential on the s.

17 **[Challenging]** • **Target: Computing z-scores** • **Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem**

A least-squares regression line predicting exam score (y) from homework completion percentage (x) is $\hat{y} = 38.0 + 0.55x$ with correlation $r = 0.82$. (a) Interpret the slope 0.55 in context. (b) Calculate and interpret the coefficient of determination r^2.

(a) Slope Interpretation: For each additional 1 percentage point increase in homework completion, the model predicts an average increase of 0.55 points on the exam. (b) r^2 Calculation & Interpretation: $r^2 = (0.82)^2 = 0.6724 =$ **67.24%**. Exactly 67.24% of the variability in exam scores is explained by the linear relationship with homework completion percentage.

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18 **[Challenging] • Target: Comparing values across different distributions • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing**

Level 1 · FoundationsAn airline wants to inspect passenger satisfaction. It randomly selects 10 flights departing from a major hub on a given day and surveys every passenger on those 10 flights. Identify this sampling method and distinguish it from stratified sampling.

This is a cluster sample. In cluster sampling, the population is divided into clusters (individual flights), a random sample of whole clusters is selected, and all individuals within those clusters are surveyed. In stratified sampling, individuals from every group would be sampled, whereas here, entire flights are either completely included or completely excluded.

19 **[Challenging] • Target: Effect of adding a constant to every data value • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference**

A sample dataset has $Q_1 = 48$ and $Q_3 = 76$. Using the 1.5 IQR rule: (a) Determine the exact lower and upper outlier fences. (b) If the dataset includes values of 4 and 122, determine whether each is an outlier.

(a) Outlier Fences: Calculate IQR $= Q_3 - Q_1 = 76 - 48 = 28$. Lower fence: $Q_1 - 1.5(\text{IQR}) = 48 - 1.5(28) = 48 - 42 = \mathbf{6}$. Upper fence: $Q_3 + 1.5(\text{IQR}) = 76 + 1.5(28) = 76 + 42 = \mathbf{118}$. (b) Outlier Status: Since $4 < 6$, 4 is a low outlier. Since $122 > 118$, 122 is a high outlier.

20 **[Challenging] • Target: Effect of multiplying every data value by a constant • Cross-Lesson: 1.01 Measures of Center & Spread**

In an SRS of 200 registered voters, 116 favor a new environmental regulation. (a) Check all three conditions for constructing a 95% confidence interval for the population proportion. (b) Construct the 95% confidence interval. (c) Interpret the interval in context.

(a) Condition Checks:Random: Stated as an SRS of 200 registered voters. (Satisfied).10% Condition: 200 voters is less than 10% of all registered voters ($N \geq 2,000$). (Satisfied).Large Counts: $n\hat{p} = 116 \geq 10$ and $n(1 - \hat{p}) = 84 \geq 10$. (Satisfied; normal approximation is valid).(b) 95% Confidence Interval: $\hat{p} = 116/200 = 0.58$, $z^* = 1.96$. SE $= \sqrt{\frac{0.58(0.42)}{200}} = \sqrt{\frac{0.2436}{200}} \approx 0.0349$. Margin of error: ME $= 1.96(0.0349) \approx 0.0684$. .

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Computing z-scores

A distribution has mean $\mu = 51$ and standard deviation $\sigma = 25$. Find the z-score for a value of $x = 41$.

a. -0.40 ✓

b. 0.40

c. -0.90

d. 0.10

$z = \frac{x - \mu}{\sigma} = \frac{41 - 51}{25} = -0.40$.

22 [Foundational] • Target: Comparing values across different distributions

On Test A (mean 79, standard deviation 4), a student scores 94. On Test B (mean 88, standard deviation 8), a different student scores 99. Using z-scores, which student performed better relative to their own test's distribution?

a. **Test A** ✓

b. Test B

c. Cannot be determined without more information

d. Both are exactly the same

Test A: $z = \frac{94 - 79}{4} \approx 3.75$. Test B: $z = \frac{99 - 88}{8} \approx 1.38$. Since $3.75 > 1.38$, Test A shows the stronger relative performance.

23 [Foundational] • Target: Effect of adding a constant to every data value

A data set has mean 75 and standard deviation 17. First, -4 is added to every value; then every value is multiplied by 4. Find the new mean and standard deviation as an ordered pair (mean, SD).

a. $(284, 17)$

b. $(285, 68)$

c. $(284, 68)$ ✓

d. $(300, 68)$

After adding -4: mean $= 71$, SD stays 17. After multiplying by 4: new mean $= 4 \times 71 = 284$, new SD $= 4 \times 17 = 68$. $(284, 68)$.

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24 [Foundational] • Target: Effect of multiplying every data value by a constant

A distribution has mean $\mu = 97$ and standard deviation $\sigma = 4$. Find the z-scores for values of 113 and 84, as an ordered pair (z_{113}, z_{84}), to two decimal places.

a. $(4.30, -3.25)$

b. $(4.00, -3.25)$ ✓

c. $(4.00, -3.55)$

d. $(-3.25, 4.00)$

$z_{113} = \frac{113-97}{4} \approx 4.00$. $z_{84} = \frac{84-97}{4} \approx -3.25$. $(4.00, -3.25)$.

25 [Foundational] • Target: Visualizing a standardized (z-score) distribution

A team of statisticians is designing a complex experiment to evaluate the efficacy of a new fertilizer on crop yield across multiple regions and soil types. They have identified 3 distinct geographic regions and 5 different soil types. Within each region-soil type combination, they plan to test 4 different concentrations of the fertilizer. To account for variability, they will run 2 independent replicates of each treatment combination. However, due to resource limitations, they can only afford to run 150 unique treatment combinations in total. The team decides to allocate the available resources by prioritizing the combinations that are most likely to reveal significant differences, while ensuring a balanced representation across the major factors. They will select a subset of the possible treatment combinations. If they decide to use a stratified sampling approach where each region is a stratum, and within each region, they stratify by soil type, and then sample fertilizer concentrations, what is the minimum number of unique treatment combinations (region, soil type, concentration) they *must* include in their sample to ensure that every soil type is represented in at least one region's stratum, and every fertilizer concentration is represented within each soil type stratum across all regions, given that they still wish to have 2 replicates per selected combination?

a. 150

b. 90

c. 75

d. **60** ✓

The experiment has 3 regions, 5 soil types, and 4 fertilizer concentrations. This gives a total of $3 \times 5 \times 4 = 60$ unique treatment combinations (region, soil type, concentration). If they run 2 replicates for each of these combinations, they would need $60 \times 2 = 120$ experimental runs. The problem states they can afford 150 unique treatment combinations in total. However, the question asks for the minimum number of *unique treatment combinations* they must include to ensure representation. To ensur.

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26 [Foundational] • Target: Applying linear transformations in context

A discrete random variable X can take on the values 1, 2, 3, and 4. The probability distribution for X is given by P(X=x) = kx, where k is a constant. What is the value of P(X=2)?

a. $\frac{1}{6}$

b. $\frac{1}{3}$ ✓

c. $\frac{1}{2}$

d. $\frac{5}{6}$

The random variable X can take on the values 1, 2, and 3. The probability distribution is given by $P(X = x) = kx$, where k is a constant. The sum of the probabilities for all possible values of X must equal 1. $P(X = 1) + P(X = 2) + P(X = 3) = 1$ Substituting the given formula for the probabilities: $k(1) + k(2) + k(3) = 1$ $k + 2k + 3k = 1$ $6k = 1$ Solving for k: $k = \frac{1}{6}$ Now we can find the probability $P(X = 2)$: $P(X = 2) = k(2)$ Substitute the value of k we found: $P(X = 2) = (\frac{1}{6})(2)$.

27 [Foundational] • Target: Finding a raw value from a z-score

A biostatistician is investigating the relationship between the dosage of a new experimental drug (measured in milligrams) and the reduction in systolic blood pressure (in mmHg) for patients with mild hypertension. They collected data from $n = 200$ patients and fit a simple linear regression model. The model's output yielded an estimated intercept of $\hat{\beta}_0 = 5.2$ and an estimated slope of $\hat{\beta}_1 = -0.8$. The standard error of the estimated slope is $\text{SE}(\hat{\beta}_1) = 0.15$. A 95% confidence interval for the true slope β_1 was calculated to be $[-1.12, -0.48]$. The biostatistician then constructs a hypothesis test to determine if the true slope is significantly different from zero. To do this, they calculate a test statistic. If the same α level used for the confidence interval were applied to a two-tailed hypothesis test of $H_0 : \beta_1 = 0$ versus $H_a : \beta_1 eq 0$, what would be the approximate p-value associated with this test statistic?

a. Greater than 0.05

b. Equal to 0.05

c. **Less than 0.05** ✓

d. Cannot be determined from the given information

The problem asks for the approximate p-value of a two-tailed hypothesis test for the slope parameter β_1 in a simple linear regression model. We are given a 95% confidence interval for β_1 as $[-1.12, -0.48]$. A confidence interval and a hypothesis test are related. For a $100(1 - \alpha)\%$ confidence interval, if the null hypothesis value (in this case, $\beta_1 = 0$) is *outside* the confidence interval, we reject H_0 at the α significance level. Conversely, if the null hypothesis value is *in.

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28 [Foundational] • Target: Combined shift-and-scale transformations

A manufacturing company produces electronic components. They are investigating potential relationships between the type of raw material used and the number of defects per batch. After collecting data for 100 batches, they create a two-way table summarizing the findings. The table shows that 15 batches using Material A had no defects, 20 batches using Material A had 1-2 defects, and 10 batches using Material A had more than 2 defects. For Material B, 25 batches had no defects, 28 batches had 1-2 defects, and 22 batches had more than 2 defects. If the company decides to perform a chi-square test for homogeneity to compare the defect rates across the two materials, what is the degrees of freedom for this test?

a. 2

b. 3

c. **4** ✓

d. 5

The problem describes a chi-square test for homogeneity, which is used to compare the distribution of a categorical variable across different populations or groups. In this case, the categorical variable is the number of defects (No Defects, 1-2 Defects, More than 2 Defects), and the populations/groups are the two types of raw materials (Material A and Material B). The structure of the data can be represented by a contingency table. The number of rows in the contingency table corresponds to the number of population.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Computing z-scores**

A research team is investigating the relationship between socioeconomic status (SES) and preferred mode of transportation for commuting to work in a large metropolitan area. They conduct a survey and categorize SES into three levels: Low, Medium, and High. The preferred mode of transportation is categorized as Public Transit, Private Vehicle, or Bicycle/Walking. After collecting data from a sample of $n = 500$ individuals, they decide to perform a chi-square test for independence. The preliminary analysis suggests that the expected counts for several cells are small. To ensure the validity of the chi-square approximation, the research team decides to merge categories. They consolidate 'Bicycle/Walking' into 'Public Transit' to form a new category 'Active/Public Transit'. If the original analysis involved 3 levels of SES and 3 modes of transportation, what is the minimum number of observations required in each cell of the *original* contingency table for the chi-square test to be considered reasonably reliable, assuming equal distribution across all cells?

a. 4

b. **5** ✓

c. 6

d. 10

The chi-square test for independence is used to analyze the association between two categorical variables. The test statistic relies on comparing observed frequencies to expected frequencies. A common rule of thumb for the validity of the chi-square approximation is that most expected cell counts should be at least 5, and no expected cell count should be less than 1. In this problem, the original contingency table has 3 levels of SES (rows) and 3 modes of transportation (columns), resulting in a 3×3 ta.

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30 **[Intermediate]** • **Target: Comparing values across different distributions**

A local government is analyzing the relationship between the socioeconomic status (SES) of residents in a city and their reported level of civic engagement. They surveyed a random sample of 500 residents and categorized them into three SES levels: Low, Medium, and High. Civic engagement was also categorized into three levels: None, Some, and High. The results are summarized in the two-way table below. A chi-square test for homogeneity is planned to investigate if the distribution of civic engagement levels differs across SES groups. | SES Level | None | Some | High | Total | |---|---|---|---|---| | Low | 60 | 90 | 50 | 200 | | Medium | 70 | 80 | 50 | 200 | | High | 30 | 40 | 30 | 100 | | Total | 160 | 210 | 130 | 500 | If the null hypothesis of homogeneity is rejected, which of the following statements best describes a potential conclusion from this test, assuming the observed counts are representative of the city's population?

a. There is no association between socioeconomic status and civic engagement.

b. Socioeconomic status has no impact on the distribution of civic engagement levels.

c. The proportion of residents with high civic engagement is the same across all socioeconomic status levels.

d. **The distribution of civic engagement levels significantly differs across the socioeconomic status groups.** ✓

The chi-square test for homogeneity tests whether the distribution of a categorical variable (civic engagement) is the same across different groups (socioeconomic status levels). If the null hypothesis is rejected, it means there is sufficient evidence to conclude that the distribution of civic engagement levels is NOT the same across all SES groups. This directly supports statement D. Statement A is too broad and refers to association, which is tested by a chi-square test of independence. While related, homogeneit.

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31 [Intermediate] • Target: Effect of adding a constant to every data value

A researcher is analyzing the relationship between the number of hours a student spends on extracurricular activities per week and their final exam score in a challenging STEM course. The researcher collects data from 100 students and fits a linear regression model. The model is given by $\text{Exam Score} = 65 + 2.5 \times \text{Hours} - 0.3 \times \text{Hours}^2$, where Exam Score is the final exam score and Hours is the number of hours spent on extracurricular activities per week. The researcher believes that for a certain range of hours, there might be an optimal number of extracurricular hours that maximizes the exam score, and beyond that, the score might decrease due to time constraints. \\ \\ Consider the function representing the exam score as a function of hours: $f(h) = 65 + 2.5h - 0.3h^2$. To find the maximum exam score and the corresponding hours, calculus can be used. \\ Which of the following is the approximate number of extracurricular hours per week that would maximize the student's final exam score, assuming the model holds for $h \geq 0$?

a. 4.17 hours ✓

b. 2.5 hours

c. 5.0 hours

d. 8.33 hours

The exam score is modeled by the quadratic function $f(h) = 65 + 2.5h - 0.3h^2$. This is a downward-opening parabola, so its maximum occurs at the vertex. To find the vertex, we can use calculus and find the critical points by taking the derivative with respect to h and setting it to zero: \\ $f'(h) = \frac{d}{dh}(65 + 2.5h - 0.3h^2) = 2.5 - 0.6h$. \\ Setting the derivative to zero to find the critical point: \\ x. \\ To confirm this is a maximum, we can check the second derivative: $f''(h) = -0.6$. Since the s.

32 [Intermediate] • Target: Effect of multiplying every data value by a constant

A political scientist is analyzing voter preferences for two candidates, Candidate A and Candidate B, across different age groups. The data is summarized in the two-way table below. The total number of surveyed individuals is $N = 500$. The total number of individuals in the 'Young Adults' row is $R_1 = 120$, and the total number of individuals who prefer Candidate A (Column 1) is $C_1 = 200$. If the political scientist were to conduct a chi-square test of independence on this data, what would be the expected count for the cell representing 'Young Adults' who prefer 'Candidate A'?

a. 48 ✓

b. 52

c. 50

d. 72

The expected count for a cell in a chi-square test of independence is calculated using the formula: $E = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$. In this case, the Grand Total $N = 500$, the Row Total for 'Young Adults' is $R_1 = 120$, and the Column Total for 'Candidate A' is $C_1 = 200$. Therefore, the expected count for the cell representing 'Young Adults' who prefer 'Candidate A' is: $E = \frac{120 \times 200}{500} = \frac{24000}{500} = 48$.

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33 **[Intermediate]** • **Target: Visualizing a standardized (z-score) distribution**

The scatterplot below displays the relationship between the number of hours (h) students spent studying for a math test and their scores (s) on that test. The equation of the least-squares regression line for this data is given by $s = 1.2h + 78.5$. If a student studied for 5 hours, what is their predicted score based on this model?

a. 91.5

b. **84.5** ✓

c. 81.5

d. 90.5

The equation of the least-squares regression line is given by $s = 1.2h + 78.5$, where s represents the predicted score and h represents the number of hours spent studying. To find the predicted score for a student who studied for 5 hours, we substitute $h = 5$ into the equation: $s = 1.2(5) + 78.5$ $s = 6.0 + 78.5$ $s = 84.5$ Therefore, the predicted score for a student who studied for 5 hours is 84.5.

34 **[Intermediate]** • **Target: Applying linear transformations in context**

The scatterplot below displays the relationship between the number of hours students spent studying for a test and their scores on that test. The equation of the least-squares regression line is approximately $\hat{y} = 55.2 + 4.8x$, where x is the number of hours spent studying and $\hat{y}$ is the predicted test score.

a. 55.2

b. **4.8** ✓

c. 50

d. 70

The slope of the least-squares regression line represents the average change in the response variable (test score) for a one-unit increase in the explanatory variable (hours studying). In the given equation $\hat{y} = 55.2 + 4.8x$, the coefficient of x is 4.8. Therefore, for each additional hour a student studies, their predicted test score increases by an average of 4.8 points.

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35 [Intermediate] • Target: Finding a raw value from a z-score

A researcher is studying the relationship between study hours and final exam scores for a group of AP Statistics students. The researcher collected data and constructed a scatterplot to visualize the relationship. The scatterplot includes a line of best fit. The equation of the line of best fit is given by $\hat{y} = 2.5x + 15$, where x represents the number of hours studied per week, and $\hat{y}$ represents the predicted final exam score. If a student studied for 12 hours, what is the predicted final exam score for that student?

a. **99** ✓

b. 45

c. 30

d. 60

The equation of the line of best fit is $\hat{y} = 2.5x + 15$. We are given that a student studied for $x = 12$ hours. To find the predicted final exam score, we substitute $x = 12$ into the equation: $\hat{y} = 2.5(12) + 15$. Calculating this gives: $\hat{y} = 30 + 15 = 45$. Therefore, the predicted final exam score is 45.

36 [Intermediate] • Target: Combined shift-and-scale transformations

The following diagram shows a scatterplot comparing the number of hours studied per week for a math course and the final exam score for a sample of students. What is the approximate y-intercept of the line of best fit?

a. 50

b. **60** ✓

c. 70

d. 80

The y-intercept of the line of best fit represents the predicted score when the number of hours studied is 0. Looking at the scatterplot, if we were to extend the apparent trend of the data back to the y-axis (where hours studied is 0), the line of best fit would cross the y-axis at approximately 60.

37 [Intermediate] • Target: Computing z-scores

A distribution is displayed as a histogram with a long tail extending to the right, most values clustered on the left. How should this shape be described?

a. **Skewed right (positively skewed)** ✓

b. Bimodal

c. Skewed left (negatively skewed)

d. Approximately symmetric (unimodal)

This description matches the shape: Skewed right (positively skewed). In this case, mean > median.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Comparing values across different distributions • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A distribution is displayed as a histogram with a long tail extending to the left, most values clustered on the right. How should this shape be described?

a. Skewed left (negatively skewed) ✓

b. Approximately symmetric (unimodal)

c. Skewed right (positively skewed)

d. Bimodal

This description matches the shape: Skewed left (negatively skewed). In this case, mean < median.

39 [Challenging] • Target: Effect of adding a constant to every data value • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

The histogram below displays the distribution of the number of hours students spent studying for a recent exam. What is the approximate median number of hours studied?

a. 2.5 hours

b. 3.5 hours ✓

c. 4.5 hours

d. 5.5 hours

The histogram shows the frequency of students studying within certain hour intervals. To find the median, we need to locate the value that divides the data into two equal halves. First, calculate the total number of students represented by the histogram. The heights of the bars represent frequencies. Let's assume the heights are as follows (this would be explicit in a real AP question with a diagram): Interval [0, 1) has frequency 5, [1, 2) has frequency 10, [2, 3) has frequency 18, [3, 4) has frequency 22, [4, 5).

40 [Challenging] • Target: Effect of multiplying every data value by a constant • Cross-Lesson: 1.01 Measures of Center & Spread

Consider a continuous random variable X with a uniform distribution over the interval $[2, 8]$. What is the probability that X is between 4 and 7, i.e., $P(4 < X < 7)$?

a. 0.50 ✓

b. 0.25

c. 0.75

d. 0.17

For a continuous uniform distribution over the interval $[a, b]$, the probability density function (PDF) is $f(x) = \frac{1}{b-a}$ for $a \le x \le b$, and 0 otherwise. In this case, $a = 2$ and $b = 8$, so the length of the interval is $b - a = 8 - 2 = 6$. The PDF is $f(x) = \frac{1}{6}$ for $2 \le x \le 8$. The probability $P(4 < X < 7)$ is the integral of the PDF from 4 to 7, or equivalently, the area under the PDF curve between 4 and 7. $P(4 < X < 7) = \int_4^7 \frac{1}{6} dx$ Since the PDF is constant, this is.

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41 [Challenging] • Target: Visualizing a standardized (z-score) distribution • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

Consider a discrete random variable X that represents the number of heads obtained when a fair coin is tossed 3 times. The probability distribution of X is given by: | x | 0 | 1 | 2 | 3 | |---------|-------|-------|-------|-------| | $P(X = x)$ | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{3}{8}$ | $\frac{1}{8}$ | What is the probability that X is less than 2, i.e., $P(X < 2)$?

a. $\frac{3}{8}$

b. $\frac{4}{8}$

c. $\frac{5}{8}$ ✓

d. $\frac{6}{8}$

The problem asks for the probability that the random variable X (number of heads in 3 coin tosses) is less than 2. This means we need to find $P(X < 2)$. The possible values of X that are less than 2 are $X = 0$ and $X = 1$. Therefore, $P(X < 2) = P(X = 0) + P(X = 1)$. From the given probability distribution: $P(X = 0) = \frac{1}{8}$ $P(X = 1) = \frac{3}{8}$ So, $P(X < 2) = \frac{1}{8} + \frac{3}{8} = \frac{4}{8}$. Simplifying the fraction, $P(X < 2) = \frac{1}{2}$. Looking at the choices provided, $\frac{4}{8}$ is equival.

42 [Challenging] • Target: Applying linear transformations in context • Cross-Lesson: 2.01 Sampling Methods & Bias

A town council is conducting a survey about a proposed new park. They decide to use a stratified random sample to ensure representation from different neighborhoods. The total population of the town is 45,000. The town is divided into three neighborhoods: Northside (18,000 residents), Eastside (15,000 residents), and Westside (12,000 residents). The council plans to survey a total of 900 residents. If the sample is to be proportional to the population of each neighborhood, how many residents should be surveyed from the Northside neighborhood?

a. **360** ✓

b. 450

c. 300

d. 400

The problem requires us to determine the number of residents to be sampled from the Northside neighborhood based on proportional allocation. 1. Calculate the proportion of the total population that resides in the Northside neighborhood: Proportion (Northside) = (Population of Northside) / (Total Population) Proportion (Northside) = 18,000 / 45,000 2. Simplify the proportion: Proportion (Northside) = 18 / 45 = 2 / 5 3. Calculate the number of residents to be sampled from the Northside by multiplying this proportion.

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43 [Challenging] • Target: Finding a raw value from a z-score • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A researcher is designing a complex experiment to investigate the impact of three different fertilizer treatments (A, B, C) on crop yield, while also controlling for two different irrigation methods (I1, I2) and two soil types (S1, S2). To ensure a robust comparison and account for potential interactions, a stratified randomized block design is employed. Within each unique combination of irrigation method and soil type (forming the blocks), the three fertilizer treatments will be randomly assigned to distinct plots. What is the total number of unique experimental units (plots) that will be established for this experiment?

a. **12** ✓

b. 6

c. 3

d. 18

The problem describes a factorial experimental design with blocking. The blocks are formed by the combinations of irrigation methods and soil types. There are 2 irrigation methods and 2 soil types, so the number of blocks is $2 \times 2 = 4$. Within each block, there are 3 fertilizer treatments that are randomly assigned to plots. Therefore, the number of experimental units (plots) per block is equal to the number of treatments, which is 3. The total number of experimental units is the number of blocks multiplied b.

44 [Challenging] • Target: Combined shift-and-scale transformations • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

An agricultural researcher is designing an experiment to study the yield of a new variety of wheat. The experiment investigates two factors: - Factor 1: Fertilizer formula, with 3 different types. - Factor 2: Watering frequency, with 4 different schedules. If the experiment uses a completely randomized design where each subject receives a combination of one fertilizer formula and one watering schedule, how many distinct treatment groups are there?

a. 7

b. 12 ✓

c. 24

d. 64

In an experiment with multiple factors, a treatment is a specific combination of levels of each factor. 1. Identify the number of levels for each factor: - Factor 1 (Fertilizer) has 3 levels. - Factor 2 (Watering) has 4 levels. 2. Apply the multiplication counting principle to find the total number of distinct treatment combinations: $\text{Number of treatments} = 3 \times 4 = 12$ - Choosing 7 is a common misconception resulting from adding the levels of the factors $(3 + 4)$ rather than multiplying them..

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45 [Challenging] • Target: Computing z-scores • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A large university is conducting a comprehensive study on student satisfaction with campus dining services. They decide to employ a complex sampling strategy involving multiple stages to ensure representation across various student demographics and living situations. The university has identified three main housing categories: On-Campus Dormitories (CD), Off-Campus Apartments (OA), and Commuters (CM). Within the On-Campus Dormitories, there are two distinct dining halls: North Campus Dining (ND) and South Campus Dining (SD). The total student population is 25,000. The university wants to select a sample that reflects the proportion of students in each housing category. Additionally, for students in On-Campus Dormitories, they want to ensure proportional representation from each dining hall. The initial allocation of the total sample size of 1000 students is as follows: 60% to On-Campus Dormitories, 30% to Off-Campus Apartments, and 10% to Commuters. For the On-Campus Dormitory sample, 70% are allocated to North Campus Dining and 30% to South Campus Dining. If the university wishes to further stratify the sample within Off-Campus Apartments based on student major (Humanities, STEM, Business), and they have determined that the proportion of Humanities majors in the Off-Campus Apartments is 0.40, STEM is 0.35, and Business is 0.25, what is the expected number of students from Off-Campus Apartments who are Humanities majors in the overall sample of 1000 students?

a. 100

b. **120** ✓

c. 140

d. 160

The total sample size is 1000 students. The proportion of the sample allocated to Off-Campus Apartments (OA) is 30%. Therefore, the number of students sampled from Off-Campus Apartments is $1000 \times 0.30 = 300$ students. Within the Off-Campus Apartments, the proportion of students who are Humanities majors is 0.40. To find the expected number of Humanities majors from Off-Campus Apartments in the overall sample, we multiply the number of students sampled from Off-Campus Apartments by the proportion of Humanities.

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